Smarter AI Agent Routing Through Task Understanding

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Background and problem statement:

Our vision is to build adaptive, self-evolving multi-agent AI systems that learn from experience and coordinate to tackle complex problems. Potential applications include developing stronger alloys, coordinating and monitoring manufacturing operations, and supporting the delivery and operation of complex infrastructure projects.

Making these systems practical requires careful use of computational resources. Using large, expensive AI models for every task can consume more money and resources than necessary. Smaller models can handle many tasks, reducing costs and dependence on major AI providers. The challenge is knowing when a smaller model is capable enough and when a task requires a larger model. A poor choice can produce unreliable answers, waste resources or cause delays, particularly when requests are ambiguous, unfamiliar or more complex than they initially appear.

Routing systems therefore need to account for uncertainty and recognise when the available information is insufficient to make a reliable selection. Within the broader research programme, this internship will focus on extracting context from task requests and test a prototype that supports better-informed routing decisions, contributing to the wider goal of balancing response quality, efficiency and cost.

Aim:

To investigate how uncertainty-aware routing can help AI systems select suitable agents and respond more effectively when a task is difficult to classify or assign.

Objectives:

- Investigate common routing failures, such as ambiguous requests and unfamiliar tasks
- Create a method to extract context from a task request.
- Develop a dataset spanning simple to complex tasks, with instructions ranging from clear and easy to understand to ambiguous and difficult to interpret.
- Explore how conformal prediction can help identify routing decisions that require additional consideration

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Expected Outcomes:

- A working prototype demonstrating uncertainty-aware agent routing
- A collection of representative tasks and challenging scenarios for evaluating routing behaviour.
- Practical evidence showing where the approach improves decisions and where it introduces additional delays or costs
- A documented codebase, final report and demonstration outlining findings and opportunities for further development.

How will this opportunity benefit you?

Skill development in the following areas:

- Applied AI development
    - Working with language models, AI agents and routing workflows
- Programming and software engineering
    - Developing, integrating, testing and documenting components using Python and version control
- Problem-solving
    - Identifying failure cases and designing appropriate responses to uncertain or unfamiliar tasks
- Experimentation and evaluation
    - Comparing approaches and assessing trade-offs between quality, speed and cost.
- Research skills
    - Reviewing relevant work, investigating ideas and drawing conclusions from evidence
- Communication and teamwork
    - Collaborating with researchers, explaining design decisions and presenting technical findings.

Other details:

- Located on campus at Burwood
- 10 weeks in duration (Unpaid)
- Undertaken as 2 days per week for a total of ~160 hours
- To be completed within the 12-week trimester period